UNIVERSITAS PADJADJARAN - Digital Business Study Program

# Statistical Analysis II - Midterm Examination (UTS) 2026/2027

Student: Tsuraya Inggit | NPM: 120510253003 | Submitted: 13 Oct 2026 | Lecturer: Dr. Vita Sarasi, S.E., M.T.

**Assigned country:** Viet Nam (last digit of NPM d = 3, 4th country). **Comparison countries:** Philippines (2nd), India (3rd). Role: business analyst advising a multinational considering investment in Viet Nam. Significance level $\alpha = 0.05$ throughout.

**Files:** this report; workbook 120510253003_TsurayaInggit_UTS.xlsx (raw downloads + transformations + live Excel formulas for Q1-Q3); reproducible code *fetch_wb.py, analysis.py, build_workbook.py, build_report.py, q7_reliability.py*; raw data in *data/raw/*. All numbers below are produced by *analysis.py* (results.json).

## *Summary of findings*

| Q | Method | Key result ($\alpha = 0.05$) | Decision / implication |
|---|---|---|---|
| 1 | Lilliefors, VN CPI inflation 1996-2025 (n=30) | D = 0.211 > 0.161; p = 0.002 -> not normal | Use quantiles/bootstrap; plan for inflation tail risk |
| 2 | $\chi^2$ 2×2, unemployed vs employed, VN vs PH, 2024 | rates 1.53% vs 2.36%; V = 0.030; weighted counts -> p unreliable | Tighter labour market in VN: hiring premium |
| 3 | $\chi^2$ 2×3 sector employment, VN vs PH | V = 0.22; industry 33% vs 18% | Manufacturing sourcing in VN; services staffing in PH |
| 4 | ANOVA / Kruskal-Wallis, GDP growth, 3 countries × 25 yrs | ANOVA p = 0.162; Kruskal-Wallis p = 0.017; Friedman p = 0.006; PH vs IN differ | VN growth ≈ India's with less than half the volatility |
| 5 | OLS GDP growth ~ investment + inflation (n=30) + quadratic | F p = 0.960; $R^2$ = 0.003; no quadratic effect | Macro predictors give no timing signal; stage investment |
| 6 | Trend vs MA(3), holdout 2021-25, exports | MA(3) MAPE 30.3% < linear 44.1%; 10-yr trend best (10.9%) | Plan on US$ 500-560 bn exports 2026-27; stage capacity |
| 7 | Reliability on public data (Zenodo, n = 248, 6 items) | $\alpha = 0.94$; Spearman-Brown = 0.95; item-total r ≥ 0.79 | Reliable but possibly redundant; validity not shown |
| 8 | Executive memo | see Q8 | REDESIGN: phased expansion with triggers |

## *Data sources, all extracted 12 Oct 2026*

| Dataset | Exact URL / API call | Indicator code | Unit | Period |
|---|---|---|---|---|
| World Bank WDI - CPI inflation | https://api.worldbank.org/v2/country/VNM;PHL;IND/indicator/FP.CPI.TOTL.ZG?format=json&per;_page=20000&date;=1960:2025 | FP.CPI.TOTL.ZG | % y/y | 1996-2025 |
| World Bank WDI - GDP growth | same, indicator NY.GDP.MKTP.KD.ZG | NY.GDP.MKTP.KD.ZG | % y/y | 2000-2024 / 1996-2025 |
| World Bank WDI - Gross capital formation | same, indicator NE.GDI.TOTL.ZS | NE.GDI.TOTL.ZS | % of GDP | 1996-2025 |
| World Bank WDI - Exports G&S | same, indicator NE.EXP.GNFS.CD | NE.EXP.GNFS.CD | current US$ | 1986-2025 |
| World Bank WDI - context (Q8) | same, indicators NE.EXP.GNFS.ZS, EG.USE.ELEC.KH.PC | NE.EXP.GNFS.ZS; EG.USE.ELEC.KH.PC | % GDP; kWh/cap | to 2025 / 2024 |
| ILOSTAT SDMX - unemployed | https://sdmx.ilo.org/rest/data/ILO,DF_UNE_TUNE_SEX_AGE_NB/VNM+PHL+IND....?format=csv&startPeriod;=2015 | UNE_TUNE_NB, age 15+ | thousand persons | 2024 |
| Zenodo - survey B (Q7) | https://zenodo.org/api/records/22057938/files/DATAUPDATE.csv/content | SCR1-SCR6 | Likert 1-5 | 2025-26 survey |
| Zenodo - survey A (Q7) | https://zenodo.org/api/records/20439760/files/Firm%20Survival%20Dataset.csv/content | OR1-OR5 | Likert 1-7 | 2025 survey |
| ILOSTAT SDMX - employed by sector | https://sdmx.ilo.org/rest/data/ILO,DF_EMP_TEMP_SEX_ECO_NB/VNM+PHL+IND....?format=csv&startPeriod;=2015 | EMP_TEMP_NB by ECO_SECTOR_* | thousand persons | 2024 |

**Data-integrity note.** Nothing was typed in or invented: World Bank JSON and ILOSTAT CSV files were downloaded programmatically and are stored unchanged in data/raw/. Viet Nam's CPI series starts in 1996, and the World Bank 2025 values are the latest available (database update 2026-10-08) and may be revised. ILOSTAT values are **weighted labour-force-survey estimates** (not raw respondent counts), which matters for Q2-Q3.

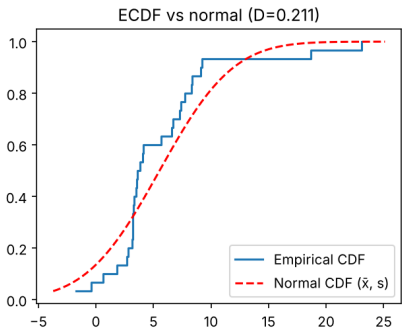
# Question 1 - Normality testing (Lilliefors)

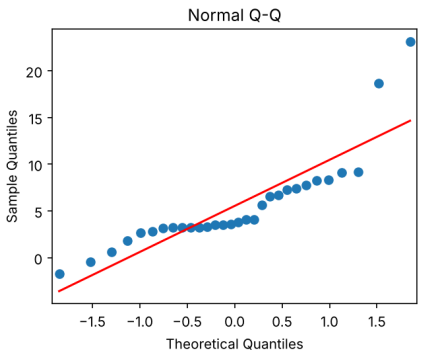
**Data.** Viet Nam CPI inflation (FP.CPI.TOTL.ZG), annual %, 1996-2025, n = 30 (≥ 20 required). Mean = 5.53, s = 5.02, median 3.73, min -1.71, max 23.12 (2008), skewness 2.04, excess kurtosis 5.44.

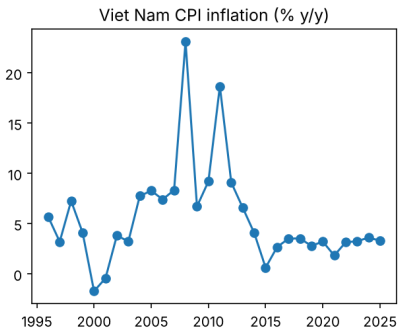
**Hypotheses.** H0: inflation is drawn from a normal distribution (mean and variance unknown, estimated from the sample); H1: it is not. **Test.** Lilliefors (1967) = Kolmogorov-Smirnov with parameters estimated: $D = \max_i \max\{ i/n - F(z_i), F(z_i) - (i-1)/n \}$, $z_i = (x_{(i)} - \text{mean})/s$, F = standard-normal CDF. Ordered observations, z-values, theoretical and empirical CDFs are in workbook sheet *Q1* (live formulas) and tables/Q1_lilliefors.csv.

| i | x(i) % | Year | z | F theor. | i/n | D+ | D- | | i | x(i) % | Year | z | F theor. | i/n | D+ | D- |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | -1.71 | 2000 | -1.44 | 0.074 | 0.033 | -0.041 | 0.074 | | 16 | 3.83 | 2002 | -0.34 | 0.367 | 0.533 | 0.166 | -0.133 |
| 2 | -0.43 | 2001 | -1.19 | 0.117 | 0.067 | -0.050 | 0.084 | | 17 | 4.08 | 2014 | -0.29 | 0.386 | 0.567 | 0.180 | -0.147 |
| 3 | 0.63 | 2015 | -0.98 | 0.164 | 0.100 | -0.064 | 0.097 | | 18 | 4.12 | 1999 | -0.28 | 0.389 | 0.600 | 0.211 | -0.178 |
| 4 | 1.83 | 2021 | -0.74 | 0.230 | 0.133 | -0.097 | 0.130 | | 19 | 5.67 | 1996 | 0.03 | 0.511 | 0.633 | 0.122 | -0.089 |
| 5 | 2.67 | 2016 | -0.57 | 0.284 | 0.167 | -0.117 | 0.151 | | 20 | 6.59 | 2013 | 0.21 | 0.584 | 0.667 | 0.083 | -0.050 |
| 6 | 2.80 | 2019 | -0.55 | 0.293 | 0.200 | -0.093 | 0.126 | | 21 | 6.72 | 2009 | 0.24 | 0.593 | 0.700 | 0.107 | -0.073 |
| 7 | 3.16 | 2022 | -0.47 | 0.318 | 0.233 | -0.084 | 0.118 | | 22 | 7.27 | 1998 | 0.35 | 0.635 | 0.733 | 0.098 | -0.065 |
| 8 | 3.21 | 1997 | -0.46 | 0.321 | 0.267 | -0.055 | 0.088 | | 23 | 7.42 | 2006 | 0.38 | 0.646 | 0.767 | 0.120 | -0.087 |
| 9 | 3.22 | 2020 | -0.46 | 0.322 | 0.300 | -0.022 | 0.056 | | 24 | 7.75 | 2004 | 0.44 | 0.671 | 0.800 | 0.129 | -0.096 |
| 10 | 3.23 | 2003 | -0.46 | 0.323 | 0.333 | 0.010 | 0.023 | | 25 | 8.28 | 2005 | 0.55 | 0.708 | 0.833 | 0.125 | -0.092 |
| 11 | 3.25 | 2023 | -0.45 | 0.325 | 0.367 | 0.042 | -0.009 | | 26 | 8.34 | 2007 | 0.56 | 0.712 | 0.867 | 0.154 | -0.121 |
| 12 | 3.31 | 2025 | -0.44 | 0.329 | 0.400 | 0.071 | -0.038 | | 27 | 9.09 | 2012 | 0.71 | 0.761 | 0.900 | 0.139 | -0.106 |
| 13 | 3.52 | 2017 | -0.40 | 0.344 | 0.433 | 0.089 | -0.056 | | 28 | 9.21 | 2010 | 0.73 | 0.768 | 0.933 | 0.165 | -0.132 |
| 14 | 3.54 | 2018 | -0.40 | 0.345 | 0.467 | 0.121 | -0.088 | | 29 | 18.68 | 2011 | 2.62 | 0.996 | 0.967 | -0.029 | 0.062 |
| 15 | 3.62 | 2024 | -0.38 | 0.351 | 0.500 | 0.149 | -0.115 | | 30 | 23.12 | 2008 | 3.51 | 1.000 | 1.000 | 0.000 | 0.033 |

| Statistic | Value |
|---|---|
| D (max \|F_n - F\|) | 0.2113 |
| Critical value, α = 0.05, n = 30 (Lilliefors 1967 table) | 0.161 |
| Approx. p-value (statsmodels, Lilliefors table method) | 0.002 |
| Decision | D > critical value and p < 0.05 -> **reject H0** |
| Cross-checks: Shapiro-Wilk (1965) / Jarque-Bera (1980) | W = 0.795, p = < 0.001 / JB = 42.7, p = < 0.001 |
| Robustness: Lilliefors on log(1+π) | D = 0.208, p = 0.002 (still rejected) |
| Lag-1 autocorrelation | 0.44 |




**Interpretation.** Normality is **not reasonable** for Viet Nam's inflation: the empirical CDF departs from the normal CDF by D = 0.211, well above 0.161. The cause is visible: a heavy right tail (skewness 2.0) created by the 2008 spike (23.1%) and 2011 (18.7%), versus a calm 2015-2025 (0.6-3.6%). Log-transforming does not fix it. **Time-series caveat:** the test

assumes independent observations; inflation is positively autocorrelated (lag-1 r = 0.44), so the effective sample size is smaller than 30 and the stated p-value is only indicative (the table critical value is for iid data). Also, the 1996-2025 span mixes policy regimes, so 'one distribution' is itself questionable.

**Implications.** (i) Do not use mean ± 1.96·s to describe 'normal' inflation ranges or to build interval forecasts for inflation; use the median/quantiles or a bootstrap. (ii) Tests on means (t, ANOVA) applied directly to inflation levels are unreliable in small samples; prefer rank-based tests or transformations and model residuals, not the raw series. (iii) For budgeting, treat inflation as skewed with tail risk (price-escalation clauses) rather than symmetric noise.

# Question 2 - Chi-square test of two proportions

**Data and outcome.** ILOSTAT, 2024, persons aged 15+: unemployed (UNE_TUNE_NB) and employed (EMP_TEMP_NB, ECO_SECTOR_TOTAL), Viet Nam vs Philippines (counts = ILO thousands × 1000). Binary outcome: unemployed vs employed within the labour force. **H0:** the unemployment proportion is equal in the two countries ($p_{VN} = p_{PH}$); H1: they differ.

| **Observed** | **Unemployed** | **Employed** | **Total** | Unemployment rate |
|---|---|---|---|---|
| Viet Nam | 858,467 | 55,280,158 | 56,138,625 | 1.53% |
| Philippines | 1,147,686 | 47,524,318 | 48,672,004 | 2.36% |

| Expected = row total × col total / N | **Unemployed** | **Employed** |
|---|---|---|
| Viet Nam | 1,074,534.8 | 55,064,090.2 |
| Philippines | 931,618.2 | 47,740,385.8 |

**Result.** $\chi^2 = \Sigma(O-E)^2/E$ = 95,384.8, df = (2-1)(2-1) = 1, p < 0.001 (Yates-corrected $\chi^2$ = 95,384.3); all expected counts > 5. Effect size: Cramér's (1946) V = 0.030 (very small); difference in rates = -0.83 pp.

**Why this p-value must not be taken at face value.** These counts are *weighted survey estimates* grossed up to national populations (N ≈ 105 million), not the number of people actually interviewed. A chi-square test treats each count as an independent observation, so with ~105 million 'observations' it will flag even trivial differences, and it ignores the survey's clustering, stratification and weighting (design effect). Unweighted respondent counts were not available from ILOSTAT; the correct approach is a survey-adjusted (Rao-Scott, 1981) test on microdata. As a transparent sensitivity analysis I re-ran the test keeping the observed proportions but assuming **illustrative** effective sample sizes per country (these n are assumptions, not data):

| Effective n per country (assumed) | $\chi^2$ | **p-value** |
|---|---|---|
| 2,000 | 3.60 | 0.058 |
| 10,000 | 18.02 | < 0.001 |
| 50,000 | 90.11 | < 0.001 |
| 200,000 | 360.43 | < 0.001 |

**Interpretation.** The *direction* (Viet Nam lower, about 0.8 pp) is stable, but whether it is statistically significant depends on the unknown effective sample size (non-significant at $n_{eff}$ = 2,000, significant from ~10,000). I therefore conclude that Viet Nam's measured unemployment rate is lower than the Philippines' in 2024, with a small practical effect, and that formal significance cannot be established from these aggregates.

**Management implication.** Both labour markets are tight by ILO's definition, but Viet Nam's lower unemployment suggests a smaller pool of immediately available workers and more wage/retention pressure than the Philippines. If expansion needs rapid hiring of non-specialised staff, budget a recruitment premium and training pipeline in Viet Nam; the Philippines offers more slack. **Limitations:** unemployment omits informality and underemployment; one year; different survey designs.

# Question 3 - Chi-square test of independence (2 × 3)

**Data.** ILOSTAT EMP_TEMP_NB by broad sector (ECO_SECTOR_AGR / IND / SER), 2024, both sexes, Viet Nam vs Philippines. **H0:** the sectoral distribution of employment is independent of country; H1: it is not. Viet Nam has 3,845,332 workers (6.96%) 'not classified by sector' (ECO_SECTOR_X), excluded from the main test; a 2×4 version including them is given as robustness.

| Observed (persons) | **Agriculture** | **Industry** | **Services** | **Total** |
|---|---|---|---|---|
| Viet Nam | 13,637,057 | 17,131,226 | 20,666,543 | 51,434,826 |
| Philippines | 9,858,510 | 8,649,772 | 29,016,036 | 47,524,318 |

| Expected under H0 | **Agriculture** | **Industry** | **Services** |
|---|---|---|---|
| Viet Nam | 12,212,013 | 13,399,885 | 25,822,928 |
| Philippines | 11,283,554 | 12,381,113 | 23,859,651 |
| Row % Viet Nam / Philippines | 26.5% / 20.7% | 33.3% / 18.2% | 40.2% / 61.1% |
| Std. residuals VN | 408 | 1,019 | -1,015 |

**Result.** $\chi^2$ = 4,653,827, df = (2-1)(3-1) = 2, p < 0.001; Cramér's V = 0.217 (moderate). Robustness 2×4 incl. 'not classified': $\chi^2$ = 8,107,444, df = 3, V = 0.281 - same conclusion. **Small cells:** minimum expected count = 11,283,554 persons - no small-cell problem (rule: all E ≥ 5). The goodness-of-fit alternative (Viet Nam counts vs Philippine shares as 'benchmark') is also rejected ($\chi^2$ = 10,944,882), but I treat that only as a descriptive check because the Philippines is not a justified published benchmark.

**Interpretation.** Sector mix clearly depends on country: Viet Nam is far more industry-oriented (33.3% vs 18.2% of classified jobs) and more agricultural (26.5% vs 20.7%), whereas the Philippines is services-dominated (61.1% vs 40.2%). As in Q2, the p-value is mechanically ~0 because the counts are weighted national estimates; the **effect size** (V ≈ 0.22, standardised residuals above ±400) is the informative quantity, and the pattern is consistent with well-known structural differences.

**Management decision (sourcing / staffing).** Locate labour-intensive manufacturing and light-assembly sourcing in Viet Nam, where industry employs 17.1 million workers and a manufacturing labour pool exists; place business-process and customer-service staffing in the Philippines (services share 61%). Verify local skills at city level before committing, since national shares do not describe a specific industrial zone. **Limitation:** one year, broad sectors, Viet Nam's 7% unclassified.

# Question 4 - One-way ANOVA across three countries

**Panel.** Annual GDP growth (NY.GDP.MKTP.KD.ZG, %), Viet Nam, Philippines, India, matching years 2000-2024 (25 per country, N = 75; sheet *Q4_panel*). **H0:** $\mu_{VN} = \mu_{PH} = \mu_{IN}$; H1: at least one differs.

| Country | Mean | Median | SD | Shapiro-Wilk W (p) |
|---|---|---|---|---|
| VNM | 6.33 | 6.69 | 1.35 | 0.861 (0.003) |
| PHL | 4.98 | 5.71 | 3.36 | 0.595 (< 0.001) |
| IND | 6.20 | 7.21 | 3.03 | 0.721 (< 0.001) |

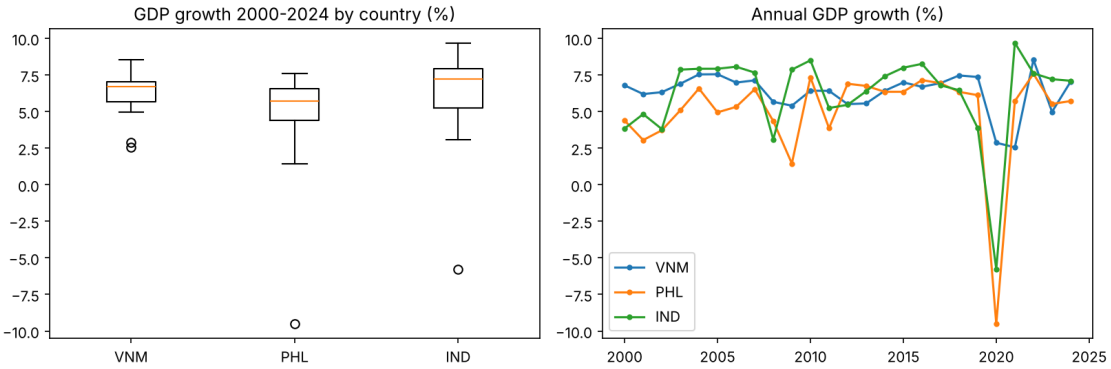
**Assumption checks.** Normality within groups is **rejected** for all three countries (Shapiro p < 0.01; skewed by crisis years such as 2020). Homogeneity of variance: Levene (1960; median-centred) F = 1.12, p = 0.333 does not reject, but Bartlett $\chi^2$ = 18.2, p = < 0.001 does, and the SDs differ visibly (1.35 vs 3.36 vs 3.03). Because Bartlett is itself sensitive to non-normality and Levene has modest power with n = 25, I treat variances as unequal and normality as violated; hence the rank-based Kruskal-Wallis test is designated the primary test, with classical and Welch ANOVA reported for completeness.

| Test | Statistic | df | p-value |
|---|---|---|---|
| Classical one-way ANOVA | F = 1.865 | 2, 72 | 0.162 |
| Welch (1951) ANOVA (unequal variances) | F = 1.710 | 2, 40.6 | 0.194 |
| **Kruskal-Wallis (1952) (primary)** | H = 8.173 | 2 | 0.017 |

ANOVA table: $SS_{between}$ = 27.76, $SS_{within}$ = 535.80, $MS_{between}$ = 13.88, $MS_{within}$ = 7.44, $\eta^2$ = 0.049.

**Post-hoc (justified because Kruskal-Wallis is significant).** Dunn's (1964) test with tie correction and Holm (1979) adjustment (rank-based, consistent with the primary test); Tukey HSD on means is shown for reference since the ANOVA F was not significant.

| Pair | Mean-rank diff. | z | p raw | p Holm | Tukey p (means) |
|---|---|---|---|---|---|
| VNM-PHL | 11.48 | 1.86 | 0.063 | 0.125 | 0.195 |
| VNM-IND | -5.84 | -0.95 | 0.343 | 0.343 | 0.986 |
| PHL-IND | -17.32 | -2.81 | 0.005 | 0.015 | 0.259 |




**Interpretation.** The classical and Welch ANOVAs find **no significant difference in mean growth** (means 6.33, 4.98, 6.20%), but the rank-based test does (mean ranks IN 45.7 > VN 39.9 > PH 28.4). I do not pick the test I like: because normality fails, Kruskal-Wallis is the appropriate one, and it indicates the distributions differ - driven by **Philippines vs India** (Holm p = 0.015); Viet Nam is not significantly different from either after Holm adjustment. Kruskal-Wallis compares distributions (location and spread), not means, so it does not prove that mean growth differs. Sensitivity: excluding 2020-2021 (COVID) the classical ANOVA becomes significant (F = 4.15, p = 0.020; means VN 6.64, PH 5.58, IN 6.57) - so the conclusion is sensitive to the pandemic years.

**Independence limitation.** Treating successive years as independent is questionable. Within-country lag-1 autocorrelation of growth is small (VN 0.08, PH -0.04, IN 0.03), but the three countries are **not independent groups**: the same years (2009, 2020) hit all of them, so the design is really repeated measures by year (a Friedman/mixed-model analysis would be more appropriate), and the 25 years are one realisation of a structurally changing economy, not

random draws. Hence p-values are indicative. As the more appropriate repeated-measures check I ran a Friedman (1937) test with the 25 years as blocks: $\chi^2(2) = 10.32$, $p = 0.006$ (mean ranks within year: VN 2.20, PH 1.48, IN 2.32), which agrees with Kruskal-Wallis: the Philippines ranks lowest in a typical year.

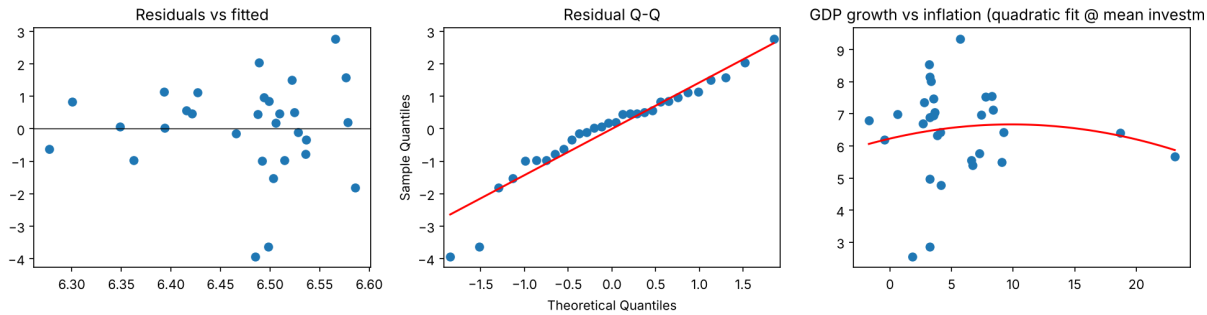
**Recommendation.** Viet Nam delivered average growth equal to India's with less than half the volatility (SD 1.35 vs 3.03 pp), which favours it for planning stability; do not claim it grows faster than India. The Philippines' growth distribution is weaker than India's.

# Question 5 - Multiple and nonlinear regression

**Data.** Viet Nam, 1996-2025, n = 30 aligned annual observations (sheet *Q5_data*). Y = GDP growth (NY.GDP.MKTP.KD.ZG, %); X1 = gross capital formation (NE.GDI.TOTL.ZS, % GDP); X2 = CPI inflation (FP.CPI.TOTL.ZG, %). Predictors were chosen a priori from growth theory (investment drives capacity; inflation proxies macro-stability/energy-cost pressure), not by searching for significance. Model: $Y = \beta 0 + \beta 1 \cdot X1 + \beta 2 \cdot X2 + \varepsilon$. H0 (each coefficient): $\beta = 0$; model F-test H0: $\beta 1 = \beta 2 = 0$.

| Term | Coef. | Std. err. | t | p-value | HAC p |
|---|---|---|---|---|---|
| Intercept | 7.212 | 3.334 | 2.16 | 0.040 | 0.032 |
| Investment (X1) | -0.022 | 0.106 | -0.20 | 0.840 | 0.830 |
| Inflation (X2) | -0.006 | 0.061 | -0.10 | 0.921 | 0.879 |

**Fit.** F(2, 27) = 0.041, p = 0.960; $R^2$ = 0.0030, adjusted $R^2$ = -0.071; residual SE = 1.50 pp. **Diagnostics.** Multicollinearity: VIF = 1.19 for both predictors (corr(X1,X2) = 0.40) - not a problem. Heteroskedasticity: Breusch-Pagan (1979) LM = 1.33, p = 0.514 (no evidence). Autocorrelation: Durbin-Watson (1950) = 1.53, Breusch-Godfrey (Breusch, 1978; Godfrey, 1978; lag 1) p = 0.376 (no evidence; HAC (Newey-West, 1987) standard errors give the same inference). Normality of residuals: Shapiro-Wilk W = 0.935, p = 0.068 (borderline pass). Largest Cook's (1977) distance = 0.18 (1996; < 1, no influential point). The diagnostics are acceptable - the problem is not a violated assumption but an **absence of signal**.




**Result.** Neither predictor is significant (p = 0.840 and 0.921), the F-test does not reject H0, and $R^2 \approx 0.003$: investment share and inflation explain essentially none of year-to-year GDP-growth variation in Viet Nam. Excluding 2020-2021 gives the same picture ($R^2$ = 0.049).

**Nonlinear extension.** Quadratic term added to inflation (centred, to limit collinearity) and, separately, to investment; compared by nested F-test, adjusted $R^2$ and AIC:

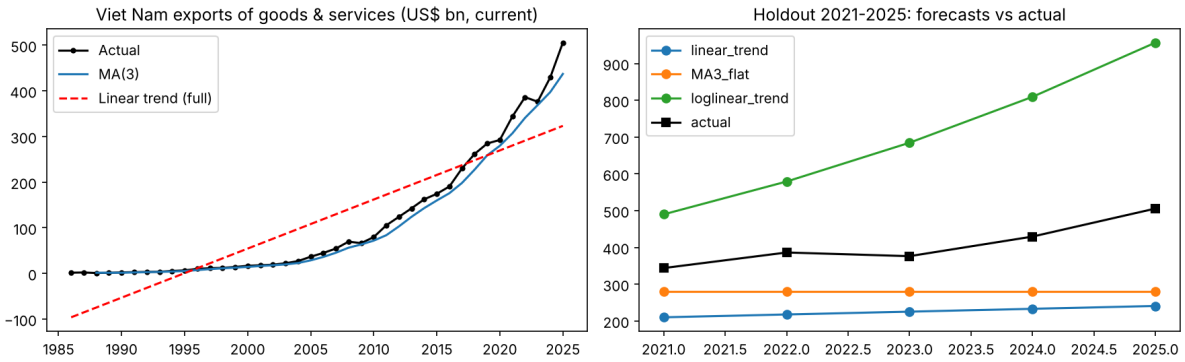
| Model | $R^2$ | Adj. $R^2$ | AIC | Quadratic term: coef (p) | Nested F-test p |
|---|---|---|---|---|---|
| Linear (X1, X2) | 0.0030 | -0.071 | 112.5 | - | - |
| + Inflation² | 0.0186 | -0.095 | 114.0 | -0.0045 (0.526) | 0.526 |
| + Investment² | 0.0199 | -0.093 | 114.0 | 0.0186 (0.510) | 0.510 |

**Interpretation.** Neither quadratic term is significant, adjusted $R^2$ falls and AIC rises: the linear model is preferred, and even it explains nothing. **Association is not causation:** even a significant coefficient here would reflect omitted variables (exports/FDI, global demand, policy, the COVID shock), reverse causality (high growth attracts investment) and the shared time trend, with n = 30 and low power. A null result also does not prove that investment or inflation do not matter, only that these two annual aggregates do not predict growth in this sample.

**Recommendation.** Do not time investment or set prices using these macro predictors. Stage the capital commitment against firm-level demand evidence, and protect margins with price-adjustment (indexation) clauses because inflation volatility (Q1) is a cost risk even if it does not predict aggregate growth in the data.

# Question 6 - Time-series forecasting

**Series.** Viet Nam exports of goods and services (NE.EXP.GNFS.CD), current US$ billion, annual, 1986-2025, n = 40 (≥ 10). It is business-relevant for supply-chain, procurement and capacity planning. **Components.** Trend: strong, rising from a few US$ bn in the 1980s to US$ 506 bn; mean annual growth 17.3% (SD 17.9 pp), i.e. closer to exponential early and decelerating later. Cycle: contractions in 2009 (-5%) and 2023 (-2.6%), slowed growth in 2020 (+3%) and a rebound in 2021 (+18%). Irregular movements: nominal-price and exchange-rate effects. **Seasonality: not identifiable** - annual data contain no within-year information, so no seasonal claim is made.




**Method.** Time-ordered holdout: models fitted on 1986-2020 and evaluated on the last 5 years (2021-2025) without shuffling. Pre-specified models: (a) linear trend y = b0 + b1·t (training slope 7.70 bn/yr); (b) 3-year moving average (multi-step forecast = last 3-year mean). Extensions added after seeing that (a) and (b) lag a still-growing series: linear trend on the last 10 training years, damped Holt trend (Holt, 1957/2004; Gardner and McKenzie, 1985), log-linear trend and a naive last-value benchmark. A rolling one-step MA(3) is also shown but is *not comparable* with multi-step forecasts.

| Model (holdout 2021-2025) | MAE (US$ bn) | **MAPE (%)** | **RMSE** |
|---|---|---|---|
| Linear trend (full training) [pre-spec.] | 182.9 | 44.1 | 188.5 |
| MA(3), 5-step flat [pre-spec.] | 128.6 | 30.3 | 140.2 |
| Linear trend, last 10 yrs [ext.] | 46.9 | 10.9 | 55.3 |
| Damped Holt [ext.] | 60.4 | 14.1 | 68.9 |
| Naive last value [ref.] | 115.8 | 27.1 | 128.6 |
| Log-linear trend [ext.] | 295.7 | 70.4 | 316.7 |
| MA(3) rolling 1-step [not comparable] | 69.6 | 16.8 | 73.6 |

MAE/MAPE follow Hyndman and Koehler (2006); MAPE is valid because the series is strictly positive. **Result:** of the two pre-specified models, the moving average beats the linear trend (MAE 128.6 vs 182.9; MAPE 30.3% vs 44.1%) - but both under-forecast badly because they lag a rising series. The extension 'linear trend on the last 10 years' is best (MAPE 10.9%); the log-linear trend over-forecasts (MAPE 70.4%) as growth slowed. With only 5 holdout points this ranking is itself uncertain, and the window length was chosen by judgement, so I treat it as a reasonable planning model, not a proven winner.

| Forecast (US$ bn) | **2026** | **2027** | 95% interval |
|---|---|---|---|
| Linear trend, full series | 334 | 345 | 192-476; 202-487 |
| MA(3) | 437 | 437 | not defined (no error model) |
| Linear trend, last 10 years (recommended base) | 502 | 533 | 447-557; 476-591 |
| Damped Holt | 518 | 558 | - |

**Uncertainty and validity warnings.** The full-series linear trend forecasts US$ 334 bn for 2026, *below the 2025 actual of 506 bn*, and the 2025 actual lies outside its own 95% interval: the model is mis-specified (Durbin-Watson = 0.08, i.e. extreme residual autocorrelation, so its OLS intervals are far too narrow) and must not be used for planning. Intervals from the 10-year fit assume a stable linear trend and iid errors with 8 degrees of freedom, so they are optimistic. Values are nominal US$, so exchange-rate and price movements, tariffs and the energy-price news in the case can move them.

**Business response (procurement / capacity).** Plan on a base case of roughly US$ 500-560 bn exports in 2026-27 (flat to +10% vs 2025): size warehouse and logistics capacity in two stages (about 80% of the base-case volume committed, the remainder as options/leases), hold 4-6 weeks of critical-input inventory, and use review triggers - if actual 2026 exports fall below the interval lower bound (about US$ 447 bn) pause the second stage; if they exceed US$ 557 bn, accelerate it.

# Question 7 - Validity and reliability of a management survey

**Construct and respondents.** Perceived supply-chain resilience to energy-price shocks (news context: IMF warning on energy shocks, 7 Oct 2026; IEA EV-transition outlook, which reshapes energy and component supply). **Target respondents:** operations, procurement and supply-chain managers (≥ 2 years in role) of manufacturing, logistics and trading firms in Viet Nam, including MNC subsidiaries and SMEs.

### *(a) Six designed items (5-point Likert: 1 = strongly disagree … 5 = strongly agree):*

| # | Item | Facet / content-validity rationale |
|---|---|---|
| 1 | My firm could keep production running if energy prices stayed sharply higher for the next six months. | Operational continuity under an energy-price shock (core of the IMF warning) |
| 2 | We have qualified alternative suppliers for our most critical inputs. | Supplier diversification (pressure to diversify supply chains) |
| 3 | We could pass at least part of higher energy or transport costs to customers without losing major accounts. | Pricing power / cost pass-through |
| 4 | Our logistics network can be rerouted quickly if a key supplier or shipping route is disrupted. | Logistics flexibility |
| 5 | Our inventory buffers of critical inputs would cover at least one month of supply disruption. | Buffer capacity |
| 6 | Management regularly monitors energy-price and supplier-risk indicators and acts on them. | Sensing and response capability |

**Content validity.** Each item maps to one mechanism in the news (energy-cost shock, supplier diversification, buffers, flexibility, response capability), all are positively worded and about the same construct (no reverse-coding needed). Planned verification before fielding: 3-5 expert ratings of relevance to compute item-CVI (Polit and Beck, 2006; retain ≥ 0.78 with ≥ 6 experts) and cognitive pretest with ~5 managers. Limitation: six items cannot cover every facet, and pricing power (item 3) is conceptually distinct from buffers (item 5), which could lower alpha.

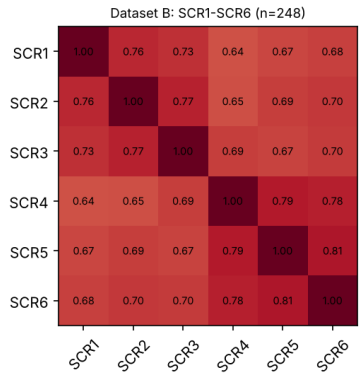
**(b) Public item-level dataset used for the computations.** I did not collect responses. I used two openly licensed (CC-BY 4.0) survey datasets from Zenodo; both are real respondent data, nothing simulated.

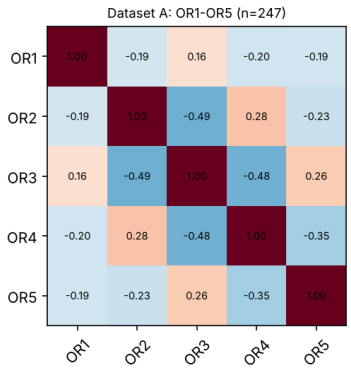
| | Dataset B (primary) | Dataset A (comparison) |
|---|---|---|
| Record | Zenodo 10.5281/zenodo.22057938 (Yuniarty, published 22 Aug 2026), file DATAUPDATE.csv - https://zenodo.org/records/22057938 | Zenodo 10.5281/zenodo.20439760 (Khan, 29 May 2026), file Firm Survival Dataset.csv - https://zenodo.org/records/20439760 |
| Respondents / scale | n = 248; 5-point Likert | n = 247 Indonesian manufacturing/supply-chain firms (managers); 7-point scale (observed 5-7) |
| Items used | SCR1-SCR6 = six-item supply-chain-resilience block (k = 6, same length as my design; exactly 3 + 3 for odd-even split) | OR1-OR5 = five 'operational resilience' items with published wording (e.g. 'Our firm can quickly recover from supply chain disruptions') |
| Why chosen | Same construct, same item count and scale type as the exam design. **Caveat:** the deposit contains no codebook or item wording (verified: record has only the CSV), so item-by-item content equivalence with my six items cannot be confirmed. | Item wording known and construct close, but only five items and a different scale. It was examined first; dataset B was selected afterwards because it matches the design better. Both are reported. |

**(c) Results.** Corrected item-total correlation $r_{i(T-i)}$ = Pearson r of item i with the sum of the other items; split-half: odd items (1,3,5) vs even items (2,4,6), Pearson $r_{half}$ then Spearman-Brown (Spearman, 1910; Brown, 1910) $r_{SB} = 2r/(1 + r)$; Cronbach's (1951) $\alpha = k/(k-1)\cdot(1 - \Sigma s^2_i/s^2_T)$, 95% CI by 2,000 bootstrap resamples.

| Item (dataset B) | **Mean** | **SD** | **% at max** | Corrected item-total r | α if deleted |
|---|---|---|---|---|---|
| SCR1 | 3.20 | 1.46 | 22.6 | 0.787 | 0.929 |
| SCR2 | 3.33 | 1.44 | 26.6 | 0.812 | 0.926 |
| SCR3 | 3.22 | 1.40 | 17.7 | 0.812 | 0.926 |
| SCR4 | 3.45 | 1.41 | 27.0 | 0.806 | 0.927 |
| SCR5 | 3.56 | 1.46 | 35.9 | 0.829 | 0.924 |
| SCR6 | 3.46 | 1.32 | 23.0 | 0.840 | 0.923 |

| Statistic (dataset B, n = 248, k = 6) | **Value** | **Benchmark** |
|---|---|---|
| Cronbach's α (95% CI) | 0.938 (0.923-0.950) | ≥ 0.70 acceptable; > 0.90 may signal redundancy |
| Mean inter-item correlation | 0.716 | 0.15-0.50 typical; higher = redundant |
| Odd-even split-half r (Pearson) | 0.908 | - |
| Spearman-Brown reliability $r_{SB}$ | 0.952 | ≥ 0.70 |
| Rank-based check: Spearman ρ half = / SB from ρ | 0.741 / 0.851 | robustness to non-linearity / ties |
| Sensitivity: α excl. 3 straight-liners / excl. 2 duplicate rows | 0.937 / 0.937 | stable |




**Interpretation (dataset B).** All corrected item-total correlations are high (0.79-0.84, far above 0.30) and deleting any item lowers or leaves α unchanged, so every item belongs to the scale. α = 0.94 and Spearman-Brown = 0.95 indicate **excellent internal consistency**; the rank-based split-half is lower (0.85) because the half-scores are bimodal (many respondents answer 1-2 or 4-5), which inflates the Pearson r. Mean inter-item r = 0.72 is above the usual 0.15-0.50 range, so some items are probably redundant and the scale could be shortened. The other three blocks in the same file also show α ≈ 0.89, which suggests a strong common response style, so high alpha is not evidence of validity.

**Dataset A (OR1-OR5) does not form a reliable scale:** α = -1.65, mean inter-item r = -0.14, all corrected item-total correlations negative (-0.48 to -0.30), odd-even r = -0.65. Positively worded items of one construct should correlate positively, and a negative α is mathematically possible only when inter-item covariances are negative; together with the restricted range (almost all answers 5-7, ceiling effects) this points to a **data-quality or coding problem** in that file. I therefore do not base conclusions on it; it is reported for transparency.

**Limitations.** (1) Reliability ≠ validity: no evidence here of convergent/discriminant validity; (2) dataset B items are not my six items and their wording is undocumented; (3) Indonesian SME/digital-business respondents, not Viet Nam; (4) Likert data are ordinal (a polychoric/ordinal alpha would be a stricter robustness check) and the sample is a convenience sample of unknown design; (5) my own six items remain *un-piloted* - the protocol above (expert CVI, pretest, ≥ 30 responses and the same script *q7_reliability.py*) is the next step before using them.

# Question 8 - Executive decision memo

**To:** Board of Directors | **Re:** Viet Nam operations | 12 October 2026

**Decision: REDESIGN.** Proceed with a phased expansion in Viet Nam, released in tranches against energy-cost and demand triggers, rather than one full commitment.

**Claim 1 (IMF, Reuters 7 Oct 2026: energy prices, record public debt and AI-investment risks threaten global growth).** This is a forward-looking warning in the news, not a Viet Nam finding, and our data cannot test an energy shock directly because no energy-price series was analysed. Historically, Viet Nam's GDP growth averaged 6.3% in 2000-2024 with the lowest volatility of the three countries (SD 1.35 pp vs Philippines 3.36, India 3.03), yet it fell to 2.9% in 2020 and 2.6% in 2021. In our regression, inflation and investment explained almost none of annual growth ($R^2 = 0.003$, $p = 0.96$), so we found no statistical evidence that domestic price pressure alone drives growth; this is association, not causation, from 30 points. Exports equal 98% of GDP (2025), so a global slowdown is the more plausible channel, which we did not model.

**Claim 2 (IEA Global EV Outlook 2026, as reported: Viet Nam's EV share of new-car sales reached about 40% in 2025, with far higher shares projected for 2035).** We cannot test sales shares with our data, and the 2035 figure is a projection, not an observation. As a loose proxy only, electricity use per capita grew 7.1% a year in 2014-2024, consistent with, but not proof of, rising power demand.

**Risk indicators.** (1) Inflation tail risk: Viet Nam's CPI inflation failed the Lilliefors normality test ($D = 0.211 > 0.161$) because of a 23.1% spike in 2008; it exceeded 4% in 14 of 30 years, although volatility since 2015 is low (SD 0.9 pp; 3.3% in 2025). (2) Export uncertainty: 2026 exports are forecast at US$ 502-518 bn with a 95% interval of roughly 447-557 bn.

**Contingency.** Stage capital in two tranches; release the second only if 2026 exports stay above US$ 447 bn and inflation below 5%; use energy-indexed supplier and customer contracts and hold 4-6 weeks of critical inputs.

**Limitations.** Annual data cannot capture within-year shocks; ILO labour figures are weighted survey estimates; and the news claims were assessed from published summaries, not the full Reuters and IEA texts.

*(Memo body word count: 365; target 300-450.)*

# References


**Data sources**

World Bank (2026). *World Development Indicators*, indicators FP.CPI.TOTL.ZG, NY.GDP.MKTP.KD.ZG, NE.GDI.TOTL.ZS, NE.EXP.GNFS.CD, NE.EXP.GNFS.ZS, EG.USE.ELEC.KH.PC; Indicators API v2, https://api.worldbank.org/v2/ (portal: https://data.worldbank.org/). Database updated 8 Oct 2026; extracted 12 Oct 2026.

International Labour Organization (2026). *ILOSTAT* SDMX API, dataflows DF_UNE_TUNE_SEX_AGE_NB and DF_EMP_TEMP_SEX_ECO_NB, https://sdmx.ilo.org/rest/ (portal: https://ilostat.ilo.org/data/). Extracted 12 Oct 2026.

Yuniarty, Y. (2026). Data for Supply Chain Risk Management Capability, Ambidexterity, and Digital Small Business Performance: The Mediating Role of Resilience [Data set]. Zenodo. https://doi.org/10.5281/zenodo.22057938 (CC-BY 4.0). Accessed 12 Oct 2026.

Khan, M. A. (2026). From Resilience to Survival: The Roles of Digital and Circular Supply Chain Capabilities in Driving Investment Efficiency [Data set]. Zenodo. https://doi.org/10.5281/zenodo.20439760 (CC-BY 4.0). Accessed 12 Oct 2026.

**News and context (claims evaluated in Q8)**

Reuters (7 Oct 2026). IMF chief warns energy shock, growing debt and AI risks threaten global growth. Original: https://www.reuters.com/world/asia-pacific/imf-chief-warns-energy-shock-growing-debt-ai-risks-threaten-global-growth-2026-10-07/ (not directly accessible to the tool; text read via syndicated copies, e.g. https://www.kitco.com/news/off-the-wire/2026-10-07/imf-chief-warns-energy-shock-growing-debt-and-ai-risks-threaten-global and https://www.thestar.com.my/business/business-news/2026/10/07/imf-chief-warns-energy-shock-growing-debt-and-ai-risks-threaten-global-growth).

International Energy Agency (20 May 2026). *Global EV Outlook 2026*. https://www.iea.org/reports/global-ev-outlook-2026/ (primary page not directly accessible to the tool; Viet Nam figures read from Viet Nam News, 'Viet Nam becomes Southeast Asia's largest EV market as sales surge', https://vietnamnews.vn/economy/1799967/viet-nam-becomes-southeast-asia-s-largest-ev-market-as-sales-surge.html).

Sarasi, V. (2026). *Statistical Analysis II - Midterm Examination (UTS), International Class 2026/2027*. Universitas Padjadjaran, Digital Business Study Program, 12 Oct 2026.

**Statistical methods**

Breusch, T. S. (1978). Testing for autocorrelation in dynamic linear models. *Australian Economic Papers, 17*(31), 334-355.

Breusch, T. S., and Pagan, A. R. (1979). A simple test for heteroscedasticity and random coefficient variation. *Econometrica, 47*(5), 1287-1294.

Brown, W. (1910). Some experimental results in the correlation of mental abilities. *British Journal of Psychology, 3*(3), 296-322.

Cook, R. D. (1977). Detection of influential observation in linear regression. *Technometrics, 19*(1), 15-18.

Cramér, H. (1946). *Mathematical Methods of Statistics*. Princeton University Press.

Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika, 16*(3), 297-334.

Dunn, O. J. (1964). Multiple comparisons using rank sums. *Technometrics, 6*(3), 241-252.

Durbin, J., and Watson, G. S. (1950). Testing for serial correlation in least squares regression I. *Biometrika, 37*(3-4), 409-428.

Friedman, M. (1937). The use of ranks to avoid the assumption of normality implicit in the analysis of variance. *Journal of the American Statistical Association, 32*(200), 675-701.

Gardner, E. S., and McKenzie, E. (1985). Forecasting trends in time series. *Management Science, 31*(10), 1237-1246.

Godfrey, L. G. (1978). Testing against general autoregressive and moving average error models when the regressors include lagged dependent variables. *Econometrica, 46*(6), 1293-1301.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics, 6*(2), 65-70.

Holt, C. C. (2004). Forecasting seasonals and trends by exponentially weighted moving averages. *International Journal of Forecasting, 20*(1), 5-10. (Originally 1957.)

Hyndman, R. J., and Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting, 22*(4), 679-688.

Jarque, C. M., and Bera, A. K. (1980). Efficient tests for normality, homoscedasticity and serial independence of regression residuals. *Economics Letters, 6*(3), 255-259.

Kruskal, W. H., and Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association, 47*(260), 583-621.

Levene, H. (1960). Robust tests for equality of variances. In I. Olkin et al. (Eds.), *Contributions to Probability and Statistics* (pp. 278-292). Stanford University Press.

Lilliefors, H. W. (1967). On the Kolmogorov-Smirnov test for normality with mean and variance unknown. *Journal of the American Statistical Association, 62*(318), 399-402.

Newey, W. K., and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica, 55*(3), 703-708.

Polit, D. F., and Beck, C. T. (2006). The content validity index: Are you sure you know what's being reported? *Research in Nursing and Health, 29*(5), 489-497.

Rao, J. N. K., and Scott, A. J. (1981). The analysis of categorical data from complex sample surveys: Chi-squared tests for goodness of fit and independence in two-way tables. *Journal of the American Statistical Association, 76*(374), 221-230.

Shapiro, S. S., and Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika, 52*(3-4), 591-611.

Spearman, C. (1910). Correlation calculated from faulty data. *British Journal of Psychology, 3*(3), 271-295.

Welch, B. L. (1951). On the comparison of several mean values: An alternative approach. *Biometrika, 38*(3-4), 330-336.

Yates, F. (1934). Contingency tables involving small numbers and the $\chi^2$ test. *Supplement to the Journal of the Royal Statistical Society, 1*(2), 217-235.

**Software**

Python 3.13.16; numpy 2.5.3; pandas 3.0.5; scipy 1.18.1; statsmodels 0.15.0; matplotlib 3.11.2; openpyxl 3.1.5; reportlab 5.0.1. Citations: Harris et al. (2020), *Nature, 585*, 357-362 (NumPy); McKinney (2010), *Proc. 9th Python in Science Conf.*, 56-61 (pandas); Virtanen et al. (2020), *Nature Methods, 17*, 261-272 (SciPy); Seabold and Perktold (2010), *Proc. 9th Python in Science Conf.*, 92-96 (statsmodels).

*Note:* reference list contains only works actually used for methods and data in this report; the news pieces are cited as sources of *claims*, not as statistical evidence.

# Appendix A - AI-use disclosure, prompts and verification

**Tool and role.** An AI coding assistant (Claude Code) was used to write the Python scripts, download the data, run the analysis, build the Excel workbook and draft this report. All numerical results are produced by the scripts in the submitted code files; none was typed in by the AI. The student remains responsible for the content.

**Prompts given to the AI (verbatim; translated from Indonesian in brackets):**

| # | Prompt | Purpose |
|---|---|---|
| 1 | "tolong kerjakan tugas di atas, dan aku ga punya datanya jadi kamu cariin" [please do the assignment above; I don't have the data so you find it] (exam PDF attached) | Task request; data to be found by the assistant |
| 2 | "npm aku 120510253003" [my student ID is 120510253003] | Country assignment (last digit 3 -> Viet Nam) |
| 3 | "kenapa ga bisa buka?" [why can't you open it?] | Network access to data portals blocked; explanation |
| 4 | "sudah" [done] | Network access enabled; work resumed |
| 5 | "buat bagus dong soalnya buat uts" [make it good, it's for the midterm] | Quality improvement (led to real public data for Q7, Friedman test, summary table) |
| 6 | "sama ga sama excel ini? kalo beda tolong benerin yg di excel samain sama yg. kamu buat" [is this Excel the same? if not, fix the Excel to match yours] (draft workbook attached) | Comparison and correction of an earlier draft workbook |
| 7 | "ubah english yg sheet dan kirim pdf dan excelnya dengan nama tsuraya inggit" [change the sheets to English and send the PDF and Excel under the name Tsuraya Inggit] | English labels and file naming |
| 8 | "coba tambahin referensi yg kamu pake di pdf nya dong dan apendix nya" [add the references you used and appendices to the PDF] | References and appendices |

**Searches run by the AI (to locate sources, not to generate data):** (i) 'IMF chief warns energy shock growing debt AI risks threaten global growth October 2026'; (ii) 'IEA Global EV Outlook 2026 electric car sales key findings Viet Nam'; (iii) 'supply chain resilience survey dataset Likert item-level ... Zenodo OR Mendeley OR Dryad OR figshare'; (iv) a search for the item wording of the Yuniarty (2026) dataset (no item wording found). Web-search summaries are secondary and must be checked against the primary pages.

**Verification already performed during preparation:** (1) data fetched programmatically from the official World Bank, ILOSTAT and Zenodo endpoints; raw responses saved unchanged (data/raw/); (2) cross-check of the Excel formulas (sheets Q1_calc, Q2_calc, Q3_calc, Q4 ANOVA, Q7_SCR_alpha) by recalculating the workbook in LibreOffice: D = 0.2113, chi-squares, ANOVA F = 1.865, $\alpha = 0.9375$ and corrected item-total correlations equal the Python output; (3) sum checks on ILO sectors (Philippines = 100%, Viet Nam 93.04% + 6.96% 'not classified'); (4) comparison of an earlier hand-transcribed draft workbook with the API data showed 8 differing GDP-growth values (rounding/vintage), so the API values are used; (5) assumption tests, robustness checks (excluding 2020-2021, straight-liners, duplicates) and the observation that the full-series linear trend forecast lies below the latest actual were reported rather than hidden.

**Verification still to be done by the student before submission:** (1) re-run *fetch_wb.py*, *analysis.py*, *build_workbook.py*, *build_report.py* and confirm identical results; (2) spot-check at least three values against the World Bank and ILOSTAT web pages; (3) read the primary Reuters and IEA pages and confirm the two claims used in Q8; (4) open the Zenodo records and confirm file contents and licence; (5) understand every test and be able to explain it. Known limits: Q2 and Q3 p-values are inflated by weighted counts; Q7 uses data from other respondents (Indonesia), and dataset B has no documented item wording.

# Appendix B - Data dictionary and transformations

| Question | Variable (code) | Raw object | Transformation / filter | n |
|---|---|---|---|---|
| Q1 | CPI inflation, VN (FP.CPI.TOTL.ZG) | data/raw/WB_FP.CPI.TOTL.ZG.json | VNM, non-missing years 1996-2025; sorted ascending; z = (x - mean)/s; no other transformation (log(1+π) only as robustness) | 30 |
| Q2 | Unemployed (ILO UNE_TUNE_NB), employed (EMP_TEMP_NB, ECO_SECTOR_TOTAL) | data/raw/DF_UNE_TUNE_SEX_AGE_NB.csv; DF_EMP_TEMP_SEX_ECO_NB.csv | REF_AREA in {VNM, PHL}; FREQ = A; TIME_PERIOD = 2024; SEX_T; AGE_YTHADULT_YGE15; value × 1,000 (UNIT_MULT = 3) | 2 × 2 |
| Q3 | Employment by sector (ECO_SECTOR_AGR / IND / SER) | data/raw/DF_EMP_TEMP_SEX_ECO_NB.csv | As Q2; ECO_SECTOR_X (not classified, VN only) excluded from main test, included in 2×4 robustness | 2 × 3 |
| Q4 | GDP growth (NY.GDP.MKTP.KD.ZG) | data/raw/WB_NY.GDP.MKTP.KD.ZG.json | VNM, PHL, IND; matching years 2000-2024 (no missing) | 75 |
| Q5 | Y = GDP growth; X1 = gross capital formation (NE.GDI.TOTL.ZS); X2 = CPI inflation | WB_NY.GDP.MKTP.KD.ZG, WB_NE.GDI.TOTL.ZS, WB_FP.CPI.TOTL.ZG | VNM; years with all three non-missing = 1996-2025; centred terms for quadratics (inflation - mean; investment - mean) | 30 |
| Q6 | Exports of goods and services (NE.EXP.GNFS.CD) | data/raw/WB_NE.EXP.GNFS.CD.json | VNM 1986-2025; divided by 10^9 (US$ bn); holdout = last 5 years; MA(3) = mean of last 3 training years | 40 |
| Q7 | SCR1-SCR6 (dataset B); OR1-OR5 (dataset A) | data/raw/zenodo_22057938_DATAUPDATE.csv (';'-separated); zenodo_20439760_FirmSurvival.csv | Items used as coded (no recoding; no reverse items); odd = items 1,3,5, even = 2,4,6; bootstrap 2,000 resamples, seed 1 | 248 / 247 |
| Q8 | Context indicators (NE.EXP.GNFS.ZS, EG.USE.ELEC.KH.PC) and results of Q1-Q6 | WB_*.json; results.json | Electricity CAGR 2014-2024 = (E2024/E2014)^(1/10) - 1 | - |

**World Bank API call pattern:** https://api.worldbank.org/v2/country/VNM;PHL;IND/indicator/<CODE>?format=json&per_page=20000&date=1960:2025. **ILOSTAT:** https://sdmx.ilo.org/rest/data/ILO,<DATAFLOW>/VNM+PHL+IND....?format=csv&startPeriod=2015. ILO values are weighted estimates in thousands of persons (UNIT_MEASURE = PS, UNIT_MULT = 3). **Data-quality notes:** Viet Nam CPI starts in 1996; Philippines has no ECO_SECTOR_X row (sectors sum to the total); 2025 World Bank values may be revised.

# Appendix C - Reproducibility

| File | Role |
|---|---|
| fetch_wb.py | Downloads World Bank series; saves raw JSON with URL and extraction date |
| analysis.py | All statistical tests; writes results.json, tables/*.csv, figures/*.png (random seed 1 for bootstrap) |
| build_workbook.py | Builds the Excel workbook (raw data, transformations, live formulas) |
| build_report.py | Builds this PDF from results.json (no number is typed in by hand) |
| q7_reliability.py | Computes corrected item-total r, odd-even Spearman-Brown and Cronbach's alpha for any 6-item CSV |
| data/raw/ | Original downloads: WB_*.json, DF_*.csv (ILOSTAT), Zenodo CSVs, ILO_SOURCES.txt |

**Run order:** *python3 fetch_wb.py; python3 analysis.py; python3 build_report.py; python3 build_workbook.py*. ILOSTAT and Zenodo files must be re-downloaded from the URLs in Appendix B and References if the raw folder is empty. Network access to api.worldbank.org, sdmx.ilo.org and zenodo.org is required for downloads.

# Appendix D - Data tables used in the analyses

## *D1. GDP growth (%), Q4 panel (and Q1/Q5 context)*

| Year | Viet Nam | Philippines | India |
|---|---|---|---|
| 2000 | 6.787 | 4.383 | 3.841 |
| 2001 | 6.193 | 3.049 | 4.824 |
| 2002 | 6.321 | 3.716 | 3.804 |
| 2003 | 6.899 | 5.087 | 7.860 |
| 2004 | 7.536 | 6.569 | 7.923 |
| 2005 | 7.547 | 4.943 | 7.923 |
| 2006 | 6.978 | 5.316 | 8.061 |
| 2007 | 7.130 | 6.519 | 7.661 |
| 2008 | 5.662 | 4.344 | 3.087 |
| 2009 | 5.398 | 1.448 | 7.862 |
| 2010 | 6.423 | 7.334 | 8.498 |
| 2011 | 6.413 | 3.858 | 5.241 |
| 2012 | 5.505 | 6.897 | 5.456 |
| 2013 | 5.554 | 6.751 | 6.386 |
| 2014 | 6.422 | 6.348 | 7.410 |
| 2015 | 6.987 | 6.348 | 7.996 |
| 2016 | 6.690 | 7.149 | 8.256 |
| 2017 | 6.940 | 6.931 | 6.795 |
| 2018 | 7.465 | 6.341 | 6.454 |
| 2019 | 7.359 | 6.119 | 3.871 |
| 2020 | 2.865 | -9.518 | -5.778 |
| 2021 | 2.554 | 5.715 | 9.690 |
| 2022 | 8.538 | 7.581 | 7.609 |
| 2023 | 4.979 | 5.519 | 7.210 |
| 2024 | 7.040 | 5.715 | 7.099 |

## *D2. Regression data, Q5 (Viet Nam)*

| Year | GDP growth Y (%) | Investment X1 (% GDP) | Inflation X2 (%) |
|---|---|---|---|
| 1996 | 9.340 | 28.103 | 5.675 |
| 1997 | 8.152 | 28.300 | 3.210 |
| 1998 | 5.764 | 29.050 | 7.266 |
| 1999 | 4.774 | 27.630 | 4.117 |
| 2000 | 6.787 | 29.610 | -1.710 |
| 2001 | 6.193 | 31.173 | -0.432 |
| 2002 | 6.321 | 33.221 | 3.831 |
| 2003 | 6.899 | 35.445 | 3.235 |
| 2004 | 7.536 | 35.465 | 7.755 |
| 2005 | 7.547 | 33.757 | 8.285 |
| 2006 | 6.978 | 34.537 | 7.418 |
| 2007 | 7.130 | 39.566 | 8.344 |
| 2008 | 5.662 | 36.493 | 23.115 |
| 2009 | 5.398 | 37.163 | 6.717 |
| 2010 | 6.423 | 37.099 | 9.207 |
| 2011 | 6.413 | 32.371 | 18.678 |
| 2012 | 5.505 | 30.559 | 9.095 |
| 2013 | 5.554 | 30.213 | 6.593 |
| 2014 | 6.422 | 30.289 | 4.085 |
| 2015 | 6.987 | 32.109 | 0.631 |
| 2016 | 6.690 | 31.725 | 2.668 |
| 2017 | 6.940 | 32.305 | 3.520 |
| 2018 | 7.465 | 32.020 | 3.540 |
| 2019 | 7.359 | 31.980 | 2.796 |
| 2020 | 2.865 | 31.916 | 3.221 |
| 2021 | 2.554 | 32.896 | 1.835 |
| 2022 | 8.538 | 32.340 | 3.157 |
| 2023 | 4.979 | 31.668 | 3.253 |
| 2024 | 7.040 | 30.585 | 3.621 |
| 2025 | 8.019 | 30.795 | 3.310 |

### *D3. CPI inflation, Q1 (Viet Nam, % y/y) and exports, Q6 (US$ bn)*

| Year | CPI inflation (%) | Exports (US$ bn) |
|---|---|---|
| 1986 | | 1.74 |
| 1987 | | 2.20 |
| 1988 | | 1.00 |
| 1989 | | 1.50 |
| 1990 | | 2.33 |
| 1991 | | 2.97 |
| 1992 | | 3.43 |
| 1993 | | 3.79 |
| 1994 | | 5.54 |
| 1995 | | 6.80 |
| 1996 | 5.675 | 10.08 |
| 1997 | 3.210 | 11.57 |
| 1998 | 7.266 | 12.20 |
| 1999 | 4.117 | 14.33 |
| 2000 | -1.710 | 16.81 |
| 2001 | -0.432 | 18.00 |
| 2002 | 3.831 | 19.19 |
| 2003 | 3.235 | 22.42 |
| 2004 | 7.755 | 27.13 |
| 2005 | 8.285 | 36.71 |
| 2006 | 7.418 | 44.94 |
| 2007 | 8.344 | 54.59 |
| 2008 | 23.115 | 69.72 |
| 2009 | 6.717 | 66.37 |
| 2010 | 9.207 | 79.75 |
| 2011 | 18.678 | 105.59 |
| 2012 | 9.095 | 124.15 |
| 2013 | 6.593 | 142.76 |
| 2014 | 4.085 | 162.48 |
| 2015 | 0.631 | 174.47 |
| 2016 | 2.668 | 190.53 |
| 2017 | 3.520 | 230.04 |
| 2018 | 3.540 | 261.80 |
| 2019 | 2.796 | 284.74 |
| 2020 | 3.221 | 292.48 |
| 2021 | 1.835 | 343.94 |
| 2022 | 3.157 | 386.24 |
| 2023 | 3.253 | 376.27 |
| 2024 | 3.621 | 429.48 |
| 2025 | 3.310 | 505.66 |

Q7 item-level responses (248 × 6) are in the workbook (sheet Q7_SCR_alpha) and data/raw/; they are not reproduced here.

# Appendix E - Decision log, deviations and known limitations

**Indicator choices.** Q1 uses CPI inflation (a recommended indicator) rather than GDP growth because inflation is central to the case context; Q4-Q5 use GDP growth and Q6 uses exports. Choices were made before looking at test results.

**Q2/Q3 data.** Only weighted ILO estimates in thousands were available; unweighted respondent counts and microdata were not obtained. Sensitivity $n_{eff}$ values in Q2 are illustrative assumptions, not data.

**Q4.** Because normality was rejected, Kruskal-Wallis was designated primary; the divergence from ANOVA/Welch is reported. The COVID-excluded sensitivity was a planned robustness check, reported even though it changes the ANOVA conclusion.

**Q5.** Predictors (investment, inflation) were fixed a priori; the null result was kept. No alternative predictor sets were searched.

**Q6.** Linear trend and MA(3) were pre-specified; the 10-year-window linear trend, damped Holt and log-linear models were added after observing that the pre-specified models lag a growing series, and are labelled extensions. The holdout has only 5 points, so model ranking is uncertain.

**Q7.** Dataset A (OR1-OR5) was examined first and gave negative reliability; dataset B (SCR1-SCR6) was then selected because it matches the six-item, 5-point design. Both are reported. Item wording for dataset B is undocumented; respondents are not from Viet Nam; the six items I designed were not administered.

**Q8.** News claims were read via secondary copies; the memo's EV statements are 'as reported' and must be checked against the IEA report.

**General.** Annual macro series are serially dependent and countries share global shocks, so p-values are indicative; association is not causation; 2025 values may be revised.